# SERIES (Ross Ch. 2, Rudin Ch. 3)

<u>Def</u> Let $\{a_n\}_{n\geq 1}$ be a sequence of real numbers.
We say:
① the series $\sum_{n=1}^{\infty} a_n$ <u>converges</u>/<u>diverges</u> if the sequence $\{s_n\}_{n\geq 1}$ of partial sums $s_n = a_1 + a_2 + \dots + a_n$ converges/diverges.
② the series $\sum_{n=1}^{\infty} a_n$ <u>converges absolutely</u> if the series $\sum_{n=1}^{\infty} |a_n|$ converges.

<u>Rmk</u> As $|a_n| \geq 0$, the sequence $s_n = |a_1| + \dots + |a_n|$ is increasing. So either $\sum_{n=1}^{\infty} |a_n|$ converges or diverges to $+\infty$.

<u>Thm</u> (Cauchy criterion) The series $\sum_{n=1}^{\infty} a_n$ converges if and only if: $\forall \varepsilon > 0$, $\exists N \in \mathbb{N}$ s.t.

$$\left| \sum_{k=m}^{n} a_k \right| < \varepsilon \qquad \forall n \geq m \geq N.$$

<u>Pf</u>: Recall:

$$\sum_{n=1}^{\infty} a_n \text{ converges} \overset{\text{def}}{\iff} \{s_n\}_{n\geq 1} \text{ converges}$$

$$\overset{\text{lecture 8}}{\iff} \{s_n\}_{n\geq 1} \text{ is Cauchy}$$

$$\iff \forall \varepsilon > 0, \exists N \in \mathbb{N} \text{ s.t. } |s_n - s_m| < \varepsilon \ \forall n, m \geq N.$$

When $n \geq m \geq N+1$,

$$s_n - s_{m-1} = \sum_{k=1}^{n} a_k - \sum_{k=1}^{m-1} a_k = \sum_{k=m}^{n} a_k$$

When $m \geq n$, swap $m$ and $n$ above. Together, we see that

$$|s_n - s_m| < \varepsilon \quad \forall n, m \geq N \iff \left| \sum_{k=m}^{n} a_k \right| < \varepsilon \quad \forall n \geq m \geq N+1.$$

$\square$

<u>Cor</u> If $\sum_{n=1}^{\infty} a_n$ converges, then $\lim_{n \to \infty} a_n = 0$.

<u>Pf</u>: Suppose $\sum_{n=1}^{\infty} a_n$ converges. By the Cauchy criterion, $\forall \varepsilon > 0 \ \exists N \in \mathbb{N}$ s.t.

$$\left| \sum_{k=m}^{n} a_k \right| < \varepsilon \quad \forall n \geq m \geq N.$$

Taking $n = m$, we see that

$$|a_m| < \varepsilon \quad \forall m \geq N.$$

So $\lim_{m \to 0} a_m = 0$. $\square$

<u>Cor</u> If $\sum_{n=1}^{\infty} a_n$ converges absolutely, then it converges.

<u>Pf</u>: Suppose $\sum_{n=1}^{\infty} a_n$ converges absolutely. Then $\sum_{n=1}^{\infty} |a_n|$ converges. Fix $\varepsilon > 0$. By the Cauchy criterion, $\exists N \in \mathbb{N}$ s.t.

$$\sum_{k=m}^{n} |a_k| < \varepsilon \quad \forall n \geq m \geq N.$$

So

$$\left| \sum_{k=m}^{n} a_k \right| \leq \sum_{k=m}^{n} |a_k| < \varepsilon \quad \forall n \geq m \geq N.$$

Therefore $\sum_{n=1}^{\infty} a_n$ converges by the Cauchy criterion. $\square$

<u>Ex</u> For which $r \in \mathbb{R}$ does the geometric series $\sum_{n=0}^{\infty} r^n$ converge?

① Case: $|r| \geq 1$. Then $|r^n| = |r|^n \geq 1$ $\forall n \geq 1$, and so $r^n \not\to 0$ as $n \to \infty$. By the first corollary, $\sum_{n=0}^{\infty} r^n$ must diverge.

② Case: $|r| < 1$. Note that

$$1 - r^{n+1} = (1-r)(1 + r + r^2 + \dots + r^n)$$

$$\Rightarrow \sum_{k=0}^{n} r^k = \frac{1-r^{n+1}}{1-r} \to \frac{1}{1-r} \quad \text{as } n \to \infty.$$

So $\sum_{n=0}^{\infty} r^n$ converges and $\sum_{n=0}^{\infty} r^n = \frac{1}{1-r}$.

<u>Thm</u> (Comparison test) Let $\sum_{n=1}^{\infty} a_n$ be a series with $a_n \geq 0$ $\forall n \geq 1$.

① If $\sum_{n=1}^{\infty} a_n$ converges and $|b_n| \leq a_n$ $\forall n \geq 1$, then $\sum_{n=1}^{\infty} b_n$ converges.

② If $\sum_{n=1}^{\infty} a_n$ diverges and $b_n \geq a_n$ $\forall n \geq 1$, then $\sum_{n=1}^{\infty} b_n$ diverges.

<u>Pf</u>: ① Fix $\varepsilon > 0$. As $\sum_{n=1}^{\infty} a_n$ converges, $\exists N \in \mathbb{N}$ s.t.

$$\left| \sum_{k=m}^{n} a_k \right| < \varepsilon \quad \forall n \geq m \geq N.$$

Then, for $n \geq m \geq N$ we have

$$\left|\sum_{k=m}^{n} b_k\right| \leq \sum_{k=m}^{n} |b_k| \leq \sum_{k=m}^{n} a_k < \varepsilon.$$

So, by the Cauchy criterion, $\sum_{n=1}^{\infty} b_n$ converges.

② As $a_n \geq 0$ $\forall n \geq 1$, the sequence $\{s_n\}_{n\geq 1}$ of partial sums $s_n = a_1 + \ldots + a_n$ is increasing. We know $\{s_n\}_{n\geq 1}$ diverges, so it must diverge to $+\infty$. Fix $M > 0$. Then $\exists N \in \mathbb{N}$ s.t.

$$a_1 + \ldots + a_n = s_n > M \quad \forall n \geq N.$$

So

$$b_1 + \ldots + b_n \geq a_1 + \ldots + a_n > M \quad \forall n \geq N.$$

Therefore the sequence $\{t_n\}_{n\geq 1}$ of partial sums $t_n = b_1 + \ldots + b_n$ also diverges to $+\infty$. So $\sum_{n=1}^{\infty} b_n$ diverges. $\square$

<u>Thm</u> (The dyadic criterion) Let $\{a_n\}_{n\geq 1}$ be a decreasing sequence with $a_n \geq 0$ $\forall n \geq 1$. The series $\sum_{n=1}^{\infty} a_n$ converges if and only if the series $\sum_{n=0}^{\infty} 2^n a_{2^n}$ converges.

<u>Ex</u> For which $\alpha \in \mathbb{R}$ does the series $\sum_{n=1}^{\infty} \frac{1}{n^\alpha}$ converge?

① Case: $\alpha \leq 0$. Then $\frac{1}{n^\alpha} = n^{-\alpha} \geq 1$ $\forall n \geq 1$.
In particular, $\frac{1}{n^\alpha} \not\to 0$ as $n \to \infty$, so $\sum_{n=1}^{\infty} \frac{1}{n^\alpha}$ must diverge.

② Case: $\alpha > 0$. Then $\{\frac{1}{n^\alpha}\}_{n\geq 1}$ is a decreasing

sequence of positive numbers. So, by the dyadic criterion,

$$\sum_{n=1}^{\infty} \frac{1}{n^\alpha} \text{ converges} \iff \sum_{n=0}^{\infty} \underbrace{2^n \frac{1}{(2^n)^\alpha}}_{= (2^{1-\alpha})^n = r^n \text{ for } r = 2^{1-\alpha}} \text{ converges}$$

$$\iff 2^{1-\alpha} < 1 \iff 1-\alpha < 0 \iff \alpha > 1.$$

Altogether, we conclude $\sum_{n=1}^{\infty} \frac{1}{n^\alpha}$ converges if and only if $\alpha > 1$.